

The students have launched a start-up company named Global3Dlabs, according to whose director, Gopal Krishna, 3D printing of food could be the next big thing. The company launched their first 3D food printer called 'Pramaan' priced at ₹ 75,000 in April with the help of a local bakery.

In USA, NASA has funded a project of 3D printing pizza for astronauts. Famous chocolate maker company Hershey has made a deal with a company from South Carolina to build 3D chocolate printer. The food industry seems to be adopting the idea of digital food.

Technology to look behind solid walls

A new technology allows robots to see through solid walls using Wi-Fi signals. University of California based scientists have been working on this project which provides X-ray vision to these robots, and they are able to look at objects and humans at the other side of thick walls by using radio frequency signals.

With this technology, the robots can not only look at the objects behind the walls but can also calculate their actual position and geometry, without any kind of previous knowledge of that area. Moreover, the robots can also identify the materials of which the objects are made of and classify them as human, metal or wood.

If a situation arises where human access is risky, the technology built

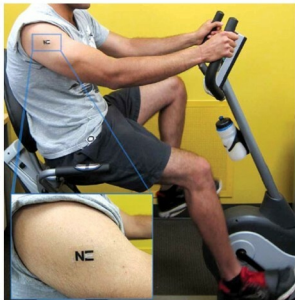
into robots can be of great help. The technology is also applicable for Wi-Fi enabled gadgets or any Wi-Fi network. When used in an existing network, it can help monitor presence and location of objects and people. This kind of monitoring helps keep a check on trespassers and even catch them.

Tattoo based bio-battery turns sweat to power

Looks like sweating it out during a daily workout has an all new meaning now, serving a dual purpose: quite useful in a multi-tasking world of ours. Pretty soon, a rigorous workout will make you fit and power your phone or other portable devices in close proximity at the same time.

A team of researchers at the University of California have successfully created a unique bio-sensor that takes the form of a temporary tattoo and uses your body sweat to monitor exercise process, all the while generating a weak electrical current. The sensor detects and interacts with lactate in sweat, a conjugate base of lactic acid that is produced by the human body via a process called glycolysis. The events gradually unfold during exercise.

Up until now, the process of monitoring and measuring the amount of lactate produced in sweat was a rather tedious process which involved taking multiple blood samples during a workout. The sensor in question is able to do away with the same, and yet is able to deliver a little



A tattoo biosensor (enlarged above) detects lactate levels during exercise; a bio-battery using the technology could power electronics (Courtesy: Joseph Wang, UCSD)

extra: generate electrical current as a by-product.

Charge your phone with background noise

Cell-phone charging will not require a charger and a socket any more. Traffic noise, music, cheering sounds from a sports ground and even your own voice can charge your phone. This innovation has been made possible by scientists from Queen Mary University of London and Nokia, who have created a prototype nanogenerator which can be used for cell-phone charging through background noises.

This energy-harvesting prototype consists of zinc oxide in the form of nanorods, which converts energy from motion into electrical energy and creates a voltage. The nanorods are coated on surfaces which make the energy harvesting process more resourceful. As soon as this surface is stretched or squashed, a high voltage is generated from the nanorods. The vibration created by any kind of sound, like our voice, catches a response from these nanorods and electrical contacts on both sides are used to harvest the voltage in order to charge a phone.



Scientists at the University of California, Santa Barbara have been working to provide X-ray vision to enable robots to see objects and humans behind thick walls through the use of radio frequency signals (Courtesy: <http://ibnlive.in.com>)

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